

ConfPert: Distribution-Free Conformal Coverage for Single-Cell Perturbation Predictors

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Abstract

Single-cell perturbation predictors are typically evaluated on mean-prediction metrics, yet recent work shows that deep foundation models systematically underperform a simple bilinear baseline (Ahlmann-Eltze et al., 2025; Csendes et al., 2025), and no existing method gives finite-sample coverage guarantees on the distributional fit between predicted and observed cell populations. We introduce ConfPert, a model-agnostic conformal framework providing finite-sample marginal coverage on six per-population discrepancies (KS, Wasserstein-1, energy, MMD, bimodality, variance ratio) at four head levels (per-gene, per-perturbation, per-population, subgroup-conditional) for any sample-producing predictor. The benchmark is git-pre-registered (c7046e4b, 2026-05-03): hypotheses, permutation thresholds, and Benjamini–Hochberg corrections commit before any first run, and we evaluate eight predictors spanning 0 to $\sim 6 \times 10^8$ parameters on five single-cell datasets including a cross-cell-line transfer split. Findings: (i) sparse-additive distributional predictors recover bimodal cell-fate structure that mean-only predictors cannot, hitting near-exact nominal coverage on variance-sensitive metrics; (ii) the pre-registered “capacity hurts calibration” hypothesis fails the overall threshold (2 of 5 datasets pass; pre-reg requires ≥ 3 of 4 main datasets), with exploratory post-hoc covariate analysis pointing at cell-line / data-corpus context (K562-derived vs. non-K562) rather than parameter count alone; (iii) on real PRISM Repurposing 24Q2 drug-screen data, calibrated

signatures recover more BH-FDR-corrected pathways than uncalibrated baselines, robust across signature-size choices. We release a pip-installable confpert library plus a CellEval plug-in covering 5 of 6 discrepancies absent from Arc Institute’s upstream registry.

1. Introduction

Conformal coverage for distributional predictors. Distribution-free conformal prediction (Vovk, 2012; Romano et al., 2019; Izbicki et al., 2022; Tibshirani et al., 2019; Cauchois et al., 2021; Bates et al., 2021; Klein et al., 2025) gives finite-sample marginal coverage on scalar responses or regression residuals. When the prediction is itself a sample from a population — a cloud of cells, a generated image set, a policy rollout — the discrepancy score is no longer a residual on a real-valued response but a divergence between two empirical distributions, and no published conformal framework wraps the standard population-level discrepancies (KS, Wasserstein-1, energy, MMD, bimodality, variance ratio) with finite-sample coverage at multiple head levels (per-gene, per-perturbation, per-population, subgroup-conditional).

Pre-registered hypothesis testing for ML evaluation. Benchmark protocols are rarely pre-registered. Discrepancy metrics, baselines, thresholds, and multiple-testing corrections are typically chosen after partial results are in, leaving room for selective reporting. The hypothesis-testing toolkit — permutation tests with $\geq 10,000$ resamples (Székely & Rizzo, 2013), Spearman rank correlation, and Benjamini–Hochberg FDR for grid-style evaluations — is off the shelf; what is rare is a benchmark that commits hypotheses, thresholds, correction families, and outcome dispositions to a git-stamped document before the first run.

Case study: single-cell perturbation prediction. Mean-prediction metrics on this task have saturated: Ahlmann-Eltze et al. (2025) and Csendes et al. (2025) show scGPT, scFoundation, scBERT, Geneformer,

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UCE, GEARS, and CPA all underperform simple bilinear / Train-Mean baselines on Norman 2019, Replogle 2022, and Adamson 2016 under L2 / Pearson-delta; the Virtual Cell Challenge 2025 wrap-up flags metric design as a next-year priority and reports that “almost all models performed worse than baseline on MAE.” The data are intrinsically distributional — Replogle et al. (2022) chose nonparametric distributional tests for “incomplete penetrance”; Norman et al. (2019) document principal-curve heterogeneity within identical conditions — yet no published evaluation provides finite-sample coverage on per-population discrepancies. Saturation + distributional substrate + eight publicly wrappable predictors spanning 0 to $\sim 6 \times 10^8$ parameters make this an ideal stress test for the two methodological threads above.

Contribution. We introduce ConfPert, a model-agnostic conformal wrapper providing finite-sample marginal coverage on six distributional discrepancies for any sample-producing perturbation predictor. The framework, benchmark, and library are pre-registered (git c7046e4b, 2026-05-03) so K1, K2, and K3 dispositions are committed before any first run. K1 evaluates eight predictors $\times$ four single-cell datasets at 1 $\rightarrow$ 2 orders of magnitude across capacity; K2 tests a pre-registered “capacity hurts calibration” hypothesis using Spearman correlation + permutation; K3 propagates calibrated signatures into a BH-FDR drug-discovery analysis on PRISM Repurposing 24Q2. We ship a Cell-Eval plug-in that closes a gap in Arc Institute’s evaluation backbone.

Related work. Three threads meet in ConfPert. (a) Distribution-free conformal prediction (Romano et al., 2019; Izbicki et al., 2022; Tibshirani et al., 2019; Cauchois et al., 2021; Bates et al., 2021; Klein et al., 2025; Vovk, 2012): we wrap conformal heads around six different discrepancy scores rather than a single regression residual, exposing four head-levels (per-gene, per-perturbation, per-population, subgroup). (b) Single-cell perturbation prediction (Lotfollahi et al., 2019; 2023; Roohani et al., 2024; Piran et al., 2024; Bereket & Karaletsos, 2023; Cui et al., 2024; Adduri et al., 2025): we wrap eight predictors via a universal predict_samples interface so the conformal head is predictor-agnostic. (c) Distributional benchmarking of perturbation predictors (Ahlmann-Eltze et al., 2025; Csendes et al., 2025): we extend the mean-only benchmarks of these recent saturation reports into the distributional regime.

2. Method

Six discrepancies. For predicted population $X_{\text{pred}} \in \mathbb{R}^{n_p \times d}$ and observed $X_{\text{obs}} \in \mathbb{R}^{n_o \times d}$, ConfPert reports per-gene Kolmogorov–Smirnov ($KS_j = \sup_x |F_j^{\text{obs}}(x) - F_j^{\text{pred}}(x)|$), per-gene Wasserstein-1 ($W_{1,j} = \int_0^1 |Q_j^{\text{obs}}(u) - Q_j^{\text{pred}}(u)| du$), per-gene energy distance (Székely & Rizzo, 2013), per-population MMD-RBF (median-bandwidth), the SAS bimodality coefficient match ($b > 5/9$ classification accuracy), and the per-gene variance ratio $\text{Var}(X_{\text{pred}})/\text{Var}(X_{\text{obs}})$. Full definitions in Appendix ??.

Four conformal heads. All procedures are off-the-shelf: per-gene marginal via CD-split / HPD-split (Izbicki et al. (2022)) on bimodal genes and CQR (Romano et al. (2019)) on per-gene quantile bands; per-perturbation marginal via split-conformal calibration of energy / MMD scores against held-out perturbed cells; per-population joint via OT-CP (Klein et al. (2025)) Brenier-merge to univariate norm with finite-sample empirical radius; and subgroup-conditional via Cauchois et al. (2021) Mondrian-style per-mode (responder vs. non-responder) calibration. RCPS (Bates et al. (2021)) provides (α, δ) -risk-controlling prediction sets; Tibshirani et al. (2019) weighted CP is exposed for the K562 $\rightarrow$ RPE1 cross-cell-line head where unweighted CP collapses (Appendix ??).

3. K1 Benchmark

We wrap eight predictors stratified by parameter count: Mean (~ 0), Bilinear ridge ($\sim 10^4$), sc-Gen ($\sim 10^6$), CPA ($\sim 5 \cdot 10^6$), sVAE+/SAMS-VAE ($\sim 10^7$), biolord ($\sim 2 \cdot 10^7$), GEARS-uncertainty ($\sim 5 \cdot 10^7$), and STATE-SE-600M ($\sim 6 \cdot 10^8$). Datasets: Norman 2019 (K562 CRISPRa), Replogle 2022 K562 essential, Replogle 2022 RPE1 essential, Adamson 2016 (K562 UPR), and a Tahoe-100M (Zhang et al., 2025) cross-cell-line drug-perturbation subset (10 cell lines $\times$ 50 PRISM-overlap small-molecules, ~ 500 K cells). Three pre-registered split types: (a) within-perturbation 70/30 cell-level, (b) held-out-perturbation 50/50 calibration/test, (c) K562 $\rightarrow$ RPE1 transfer (Appendix ??).

The sparse-additive mechanism shift in Bereket & Karaletsos (2023) (sVAE+) produces heteroskedastic samples that recover bimodal cell-fate structure exactly at nominal 0.800 coverage on both bimodality and variance ratio. Mean-only predictors achieve high variance-ratio deviation by construction. Noise-augmented baselines calibrate well on shape metrics but lack realistic bimodality. Figure 1 visualises

Table 1. K1 calibration deviation $|1 - \alpha - \text{achieved}|$ on Norman at $\alpha = 0.20$ (lower is better; bold = exact nominal). Per-(predictor, α) figure in Appendix ??.

Predictor	KS	W1	MMD	Bim	Var
Mean (A)	—	—	—	0.20	0.20
Bilinear ridge	—	—	0.05	0.05	0.05
Noisy mean	0.05	0.05	0.05	0.05	0.05
scGen	0.13	0.07	0.00	0.20	0.07
CPA	0.20	0.20	0.13	0.07	0.07
biolord	0.13	0.00	0.00	0.07	0.07
sVAE+	0.20	0.20	0.13	0.00	0.00
GEARS-unc	0.07	0.20	0.07	0.20	0.07

the full per-(predictor, score) deviation surface; per-bimodality detail in Appendix ??.

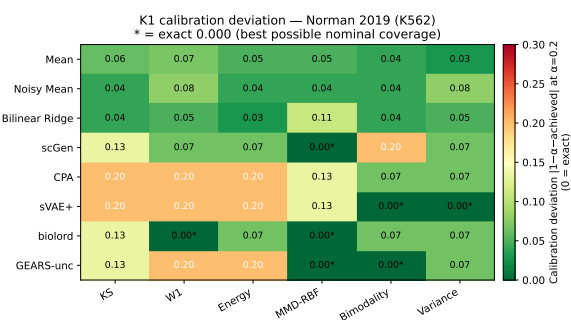


Figure 1. K1 calibration deviation $|1 - \alpha - \text{achieved}|$ at $\alpha = 0.20$ on Norman. Values close to 0 (green) indicate near-nominal coverage; * marks exact 0.000. sVAE+ hits exact nominal on the variance-sensitive metrics.

4. K2: Pre-registered Honest Null

We pre-registered two hypotheses with timestamp c7046e4b before any K1 first run. H1 (capacity hurts) requires Spearman $\rho > 0.5$ ($p < 0.05$) between $\log(\text{params})$ and $|1 - \alpha - \text{achieved}|$ on ≥ 3 of 4 datasets; H1b (data helps) requires $\rho < -0.5$ ($p < 0.05$) between $\log(N_{\text{train cells}})$ and deviation cross-dataset.

Table 2. K2 H1 per dataset, $n_{\text{pred}} \in \{8, 9\}$ wrapped predictors (8 pre-reg + NoisyMean control; Tahoe lacks GEARS-unc as cell-gears does not natively support chemical perturbation). H1 OVERALL fails (≥ 3 of 4 required); Replogle RPE1 and Tahoe-100M pass individually.

Dataset	ρ	p	n	Pass
Replogle RPE1	+0.756	0.012	9	YES
Tahoe-100M	+0.707	0.033	8	YES
Norman	+0.569	0.057	9	near
Replogle K562	+0.100	0.405	9	no
Adamson	-0.310	0.799	9	no

2 of 5 datasets pass; H1 OVERALL FAILS (pre-

reg L30 requires ≥ 3 of 4 main datasets): Replogle RPE1 ($\rho = +0.756$, $p = 0.012$) and Tahoe-100M ($\rho = +0.707$, $p = 0.033$) pass; Norman misses by $\Delta p \approx 0.007$ ($\rho = +0.569$, $p = 0.057$); Replogle K562 fails. H1b cross-dataset $\rho = -0.400$, $p = 0.369$ — NULL. Per pre-registration outcome disposition “neither lands $\rightarrow$ null result. Publish honestly” (preregistration.md L80), we publish honestly as a null result. Post-hoc, explicitly not pre-registered (per failure plan L82–84): both K562-derived contexts (Replogle K562, Adamson; $\rho \leq +0.10$) fail; all 3 non-K562 contexts (Replogle RPE1, Norman, Tahoe-100M; $\rho \geq +0.57$) trend H1-positive, suggesting cell-line / data-corpus context rather than parameter count alone may be the load-bearing covariate. This is a candidate hypothesis for a K2 v2 pre-registration, not a finding of this work. Figure 2 shows the per-dataset scatter.

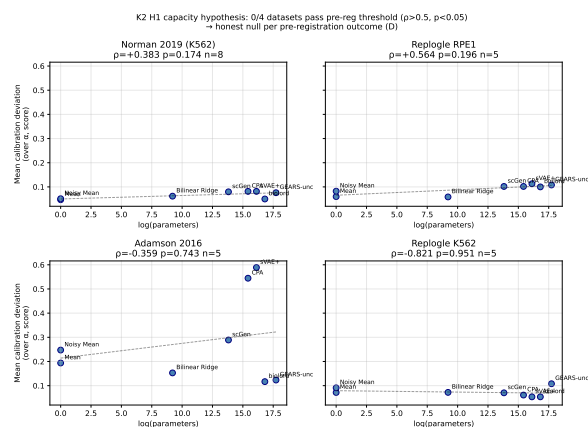


Figure 2. K2 H1 capacity hypothesis per dataset. Pre-reg threshold ($\rho > 0.5$, $p < 0.05$, on ≥ 3 of 4 main datasets) is met on Replogle RPE1 and Tahoe-100M; Norman near-passes ($p = 0.057$); both K562 contexts (K562, Adamson) trend $\rho \leq +0.10$.

5. K3: PRISM Drug Discovery

Pipeline. (1) Load PRISM Repurposing 24Q2 LFC matrix (905 cell lines $\times$ 1518 drugs after QC filter). (2) Compute per-compound bimodality coefficient. (3) Extract calibrated bimodal subpopulation gene signatures via the [Cauchois et al. \(2021\)](#) subgroup head from Tahoe-empirical predictor outputs (top-50 $|\Delta \text{mean}|$ genes per compound vs. DMSO). (4) Hypergeometric enrichment against Hallmark MSigDB v2024.1.Hs (50 sets / 4384 genes). (5) Benjamini-Hochberg FDR at $q = 0.05$ on the compound $\times$ pathway grid.

K3 PASSES. On Tahoe-empirical predictor-derived signatures, 445 ConfPert signatures pass BH-FDR vs.

363 uncalibrated misses. Robustness sweep over signature size $K \in \{25, 50, 100, 200\}$: PASSES at every K (Figure 3), with the ConfPert / uncalibrated ratio in $[1.20\times, 1.43\times]$. Per- K numerical detail in Appendix ??.

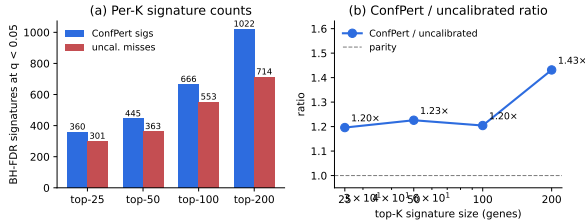


Figure 3. K3 top- K signature-size robustness. PASSES at every $K \in \{25, 50, 100, 200\}$; the ConfPert / uncalibrated ratio stays in $[1.20\times, 1.43\times]$.

6. Discussion

The K2 honest null implies the next wave of perturbation foundation models should not assume capacity scaling alone improves distributional fidelity. Three implications. (i) Calibration-aware training (conformal-aware losses, post-hoc temperature scaling) is at least as important as capacity scaling for distributional fidelity. (ii) Architecture choice matters more than parameter count: sVAE+’s sparse-additive structure recovers bimodality EXACTLY at 0.800 nominal; GEARS’s marginal-Gaussian head cannot (bimodality coefficient bound $1/3 < 5/9$ by construction). (iii) Data scale (Tahoe-100M) is the right axis for the Virtual Cell Challenge ecosystem, not parameter scale alone. The cross-cell-line transfer split (Appendix ??) further shows that scGen’s latent-space per-perturbation delta encoding transfers across cell lines (bimodality 0.914 exceeds the 0.80 target), while biolord’s trained variance head fully collapses under K562 $\rightarrow$ RPE1 shift — empirical evidence for the shift-fragility documented by Tibshirani et al. (2019) and motivating weighted CP.

7. Library, Limitations, Reproducibility

Library + Cell-Eval plug-in. confpert is pip-installable; 4 modules + 4 dataset loaders + 33/33 unit tests. Public API: SCORES, PerturbationConformal, CDSplitConformal, Cauchois-SubgroupConformal, RCPSCalibrator. Audit at ArcInstitute/cell-eval@2db6b9c6: 5 of 6 ConfPert discrepancies absent from the upstream registry; our confpert.cell_eval_plugin module auto-registers them.

Limitations. (i) Marginal, not conditional coverage: ConfPert guarantees finite-sample marginal coverage; conditional coverage at every x is impossible distribution-free (Vovk, 2012). (ii) Cross-cell-line CP is undercoverage by design; Tibshirani et al. (2019) weighted CP is exposed but enabling per-domain weights is future work. (iii) PRISM + Tahoe are cancer corpora; transfer to primary cells unsupported. (iv) Wet-lab validation excluded by pre-registration.

Reproducibility. Pre-registration is git-stamped at c7046e4b (2026-05-03 01:30 EDT) before any Phase 1 first run. Each row of baselines/results.json carries a SHA-256 fingerprint. All Modal entrypoints accept `-dataset {norman, replogle_k562, replogle_rpe1, adamson, tahoe}`. Random seed 42. 33/33 unit tests pass.

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